IS1105 — Business Information Systems

Full Marking Scheme & Model Answers · Section A

Module Code	IS1105
Examination Session	Winter 2024
Paper Number	1
Institution	University College Cork
Section Covered	Section A (Questions 1 and 2 — 50 points)

Prepared by Study-Uni. This scheme covers Section A of the Winter 2024 Paper 1 exam. Section B (25 multiple-choice questions marked via EDPAC sheet) is out of scope for this document. Model answers are provided for both Section A questions, since students may select either one.

Question 1(a) — ERP Systems

25%

QUESTION AS SET

Databases are important resources for managing an organisation's data. Critically discuss the key characteristics of different types of databases for processing transactions, including how data is stored and processed.

a) Describe how ERP systems streamline business processes and improve operational efficiency. Provide examples to illustrate your answer. (25 percent)

FULL MODEL ANSWER

What an ERP system is

An **Enterprise Resource Planning (ERP) system** is a single, integrated software platform built around a shared central database that connects the core business processes of an organisation — finance, human resources, procurement, manufacturing, inventory and sales — which would otherwise run as separate, disconnected departmental applications ("information silos"). Because every module reads and writes to the same underlying data, a business event captured once (e.g. a sales order) is immediately visible to every function that needs it, without being re-entered.

How ERP streamlines business processes

- **End-to-end process integration, not departmental hand-offs.** ERP is organised around complete business processes rather than isolated departments. The *order-to-cash* process (sales order → credit check → pick/pack → shipment → invoice → payment) and the *procure-to-pay* process (purchase requisition → purchase order → goods receipt → supplier invoice → payment) each run as one continuous, automated workflow inside the ERP instead of as separate steps between disconnected systems that require manual re-entry and reconciliation at every hand-off.
- **Elimination of duplicate data entry.** A transaction is captured once at its source and propagates automatically to every module that depends on it — removing the manual re-keying and reconciliation errors that arise when finance, sales and the warehouse each maintain their own separate records of the same event.
- **Standardised procedures across the organisation.** Because every site or department works from the same chart of accounts, the same customer and material master data, and the same approval workflows, the organisation processes transactions the same way everywhere, which simplifies training, control and audit.

How ERP improves operational efficiency

- **Faster cycle times.** Automating routine transactions (invoice matching, stock replenishment triggers, payroll runs) removes manual processing steps and shortens the time from order to cash or from requisition to payment.
- **Real-time, organisation-wide visibility.** Because all modules share one database, managers can see current stock levels, order status and financial position at any moment, rather than waiting for an end-of-month consolidation of separate systems — enabling faster, better-informed operational decisions.

- **Reduced overhead and error rates.** Removing duplicate data entry and manual reconciliation reduces both the administrative headcount required for routine processing and the error rate associated with re-keying the same data multiple times.

Illustrative example

Consider a manufacturing firm using an ERP platform such as SAP S/4HANA or Oracle NetSuite. When a customer order is entered, the system automatically checks stock availability and customer credit, reserves inventory, triggers a production or replenishment order if stock is insufficient, updates the general ledger with the expected receivable, and later reconciles the shipment and invoice — all from that single order record. Before ERP, each of these steps might have been handled in a separate sales system, warehouse system and finance system, with staff manually re-entering the same order details into each and reconciling discrepancies between them after the fact.

> **SOURCING NOTE** *The order-to-cash/procure-to-pay process framing and the SAP/Oracle NetSuite example are standard ERP-vendor and MIS-textbook content rather than a specific IS1105 lecture, since the provided course pack does not contain a dedicated ERP lecture. A student citing a different real or plausible ERP example with equivalent explanation should be credited equally.*

Mark Allocation — Question 1(a)

Band	Descriptor	Marks
Excellent *21–25*	Clearly explains how ERP's integrated, single-database design streamlines cross-functional processes (e.g. order-to-cash) **and** improves operational efficiency (cycle time, visibility, reduced error/overhead); supports the explanation with a well-developed, correctly reasoned example.	**21–25**
Good *16–20*	Explains ERP's role in streamlining processes and efficiency with reasonable detail; example given but less fully developed or only partially connects to the explanation.	**16–20**
Partial *10–15*	Basic, largely descriptive account of ERP; limited explanation of how it actually streamlines processes or improves efficiency; example, if present, is generic or unexplained.	**10–15**
Minimal *0–9*	Little or no accurate explanation of ERP; no relevant example provided.	**0–9**

Question 1(a) Total: 25% (25 marks equivalent)

Question 1(b) — Relational Databases

25%

QUESTION AS SET

b) Explain the concept of relational databases and their importance in managing business data. Provide examples of how relational databases enhance business operations and decision-making processes. (25 percent)

FULL MODEL ANSWER

The concept of a relational database

A **relational database** stores data as a set of tables (relations), each made up of rows and columns, where each table represents one type of business entity — customers, orders, products, employees. Rather than storing data in one large, duplicated file, related tables are linked through key fields: a **primary key** uniquely identifies each row in a table, and a **foreign key** in one table references the primary key of another, expressing the relationship between them (e.g. each row in an Orders table references the Customer who placed it via a customer ID). Data is queried and manipulated using **Structured Query Language (SQL)**. Well-designed relational databases are **normalised** — structured specifically to minimise redundant duplication of the same fact across multiple tables — and enforce **referential integrity**, meaning the database itself rejects, for example, an order that references a customer ID that does not exist.

Why this matters: the ACID guarantee

Relational databases are built to guarantee the **ACID properties** for every transaction: **A**tomicity (a transaction completes entirely or not at all — a funds transfer cannot debit one account without crediting the other), **C**onsistency (the database always moves from one valid state to another), **I**solation (concurrent transactions from many simultaneous users do not corrupt one another), and **D**urability (once a transaction is confirmed, it survives a subsequent system failure). This is precisely why relational databases are the standard choice for core business transaction processing — financial postings, stock movements, order processing — where the organisation cannot tolerate an inconsistent or lost record of a transaction.

Importance in managing business data

- **Reduced redundancy and improved consistency.** Because each fact (a customer's address, a product's price) is stored once and referenced by key rather than duplicated across many files, updates only need to be made in one place, removing the risk of the same fact disagreeing across different records.
- **Data integrity enforced by the system itself** — validation rules and referential integrity are enforced centrally by the Database Management System (DBMS), not left to each individual application to implement correctly.
- **Controlled, concurrent, multi-user access** — many staff and applications can safely read and write the same data at once, which high-volume transaction processing requires.

Examples of how relational databases enhance operations and decision-making

Example	How it enhances the business
Sales order processing (e.g. via a relational database underlying an ERP or CRM system)	Linking Orders, Customers and Products tables by key lets staff instantly retrieve a customer's full order history or check current stock against an order, rather than searching multiple disconnected files — directly improving order-processing speed and accuracy.
Banking transaction systems (e.g. Oracle Database, Microsoft SQL Server)	ACID-compliant transactions ensure that a funds transfer either completes fully or not at all, even if the system fails mid-transaction — a guarantee businesses depend on for trustworthy financial record-keeping.
Management reporting and decision support	Because data is consistently structured and linked by key, SQL queries can quickly aggregate and cross-tabulate data across tables (e.g. "total sales by product category by region"), giving managers reliable, accurate figures to base decisions on — directly supporting the kind of evidence-based decision-making that a DSS relies on.

CROSS-REFERENCE *See Question 2(a) in the Winter 2025 paper: this same reliable, query-able relational data is typically the source data feeding a Decision Support System's model base.*

Mark Allocation — Question 1(b)

Band	Descriptor	Marks
Excellent *21–25*	Clearly explains the relational model (tables, keys, SQL) with correct reference to integrity/ ACID properties; explicitly connects this to why relational databases matter for managing business data; gives at least one well-developed, correctly explained example of enhanced operations or decision-making.	**21–25**
Good *16–20*	Reasonable explanation of the relational concept and its business importance; at least one example given but development is less complete.	**16–20**
Partial *10–15*	Basic or partial definition of a relational database; weak or missing link to business importance; examples, if present, are generic or unexplained.	**10–15**
Minimal *0–9*	Little or no accurate explanation of relational databases; no relevant example provided.	**0–9**

Question 1(b) Total: 25% | Question 1 Total: 50 Points

Question 2(a) — Targeted Display Advertising & Big Tech Revenue

25%

QUESTION AS SET

Information Systems (IS) play a crucial role in modern business operations, but they also raise significant ethical and legal concerns, including issues related to data privacy, surveillance capitalism, and the broader "dark sides" of technology use.

a) Explain how target display advertising works and discuss its significance in the revenue models of major technology companies such as Google and Facebook. Include in your answer the role of user data as a revenue model for Big Tech. (25 percent)

FULL MODEL ANSWER

How targeted display advertising works

Targeted display advertising places visual ads (banners, video, social feed placements) in front of specific users selected by their data profile, rather than a fixed audience determined by the publication alone. The process runs, in outline, as follows:

- **Data collection.** Platforms continuously collect data on user behaviour — searches, page visits, clicks, purchases, app usage, location, and (on social platforms) declared interests and social connections — via cookies, tracking pixels, log-in data and mobile device identifiers.
- **Profiling and segmentation.** This behavioural data is aggregated into a detailed profile for each user, which the platform uses to place that user into interest, demographic and intent segments (e.g. "likely to be shopping for running shoes in the next two weeks").
- **Real-time bidding (RTB).** When a user loads a page or opens an app, an automated auction runs — typically in well under half a second — in which advertisers' automated bidding systems (Demand-Side Platforms) submit bids for the opportunity to show that specific user an ad, based on how well that user's profile matches the advertiser's target audience. The highest bidder's ad is served, and the whole process completes before the page finishes loading.

The result is that two different users viewing the same web page at the same moment can see two entirely different ads, each individually selected and priced based on what the platform knows about them.

User data as the revenue model for Big Tech

For companies such as Google (Alphabet) and Meta (Facebook/Instagram), advertising is not one revenue stream among several — it is the core business model, and user data is the raw material that makes it valuable. The more accurately a platform can predict what a specific user is interested in, the more an advertiser will pay to reach that exact user rather than a broad, untargeted audience, which is why these companies invest so heavily in collecting and analysing user data across every product they offer (search history, video-watching behaviour, social connections, location).

This dependence is visible directly in their financial results: in its most recent quarterly results, Meta generated roughly 97% of its total revenue from advertising, making it almost entirely an advertising company funded by user-data-driven targeting; Alphabet, while more diversified through Google Cloud, still derives the substantial majority of its revenue from advertising across Google Search, YouTube and its display network. Between them, Alphabet, Meta and Amazon are estimated to capture more than half of global digital advertising spend, reflecting how concentrated this data-driven advertising model has become among a small number of platforms.

> **SOURCING NOTE** *Revenue-share figures for Meta and Alphabet, and the Alphabet/Meta/Amazon combined ad-market-share estimate, are drawn from current company financial reporting and industry ad-spend forecasts (2025) rather than IS1105 course material, since exact figures shift quarter to quarter. A student citing broadly consistent figures (e.g. "the large majority" rather than an exact percentage) should not be penalised for minor variation from the specific numbers above.*

Mark Allocation — Question 2(a)

Band	Descriptor	Marks
Excellent *21–25*	Clearly explains the mechanism of targeted display advertising (data collection, profiling, real-time auction/bidding); explicitly discusses its significance to Big Tech revenue models with correct, specific reference to Google and/or Facebook/Meta; clearly explains user data's role as the underlying revenue driver.	**21–25**
Good *16–20*	Reasonable explanation of how targeted advertising works and its revenue significance; discussion of user data's role present but less fully developed.	**16–20**
Partial *10–15*	Basic or partial explanation of targeted advertising; weak or missing connection to Big Tech revenue models or the role of user data.	**10–15**
Minimal *0–9*	Little or no accurate explanation of targeted advertising or its revenue significance.	**0–9**

Question 2(a) Total: 25% (25 marks equivalent)

Question 2(b) — Data Privacy, Ethics & Surveillance Capitalism

25%

QUESTION AS SET

b) Discuss the ethical and legal concerns associated with data privacy in the context of Information Systems. In your answer, address the concept of surveillance capitalism and its implications for individual privacy and autonomy. (25 percent)

FULL MODEL ANSWER

The concept of surveillance capitalism

Surveillance capitalism is a term coined by Harvard academic Shoshana Zuboff to describe a business model in which private human experience is claimed as raw material, translated into behavioural data, and processed into "prediction products" that are sold to advertisers seeking to know what a person will do next. Unlike traditional advertising, which targets a broad audience based on where an ad is placed, surveillance capitalism depends on continuously extracting far more behavioural data than is strictly needed to deliver a service, in order to build increasingly accurate predictive profiles of individual users — directly connected to the targeted advertising mechanism discussed in Question 2(a).

Ethical concerns

- **Informed consent is often illusory.** Data collection frequently occurs through long, unread terms-of-service agreements or default-on settings, meaning users rarely give genuinely informed, specific consent to the scale of data collected about them.
- **Behavioural influence and autonomy.** Because prediction products are sold specifically to influence future behaviour (what to buy, how to vote, what to watch next), critics argue this model does not merely observe human behaviour but actively shapes it, raising a genuine ethical question about individual autonomy — the ability to make one's own choices free from manipulation.
- **Power asymmetry.** A small number of platforms accumulate a scale of behavioural data about individuals that neither the individual nor most regulators can fully see or audit, creating a significant imbalance of information and power between the platform and the individual.
- **Algorithmic bias and discrimination.** Profiles built from historical behavioural data can encode and reinforce existing societal biases, for example in ad delivery or credit-related targeting, raising fairness concerns distinct from privacy alone.

Legal concerns and the regulatory response

The primary legal framework governing this issue in the EU (and therefore in Ireland) is the **General Data Protection Regulation (GDPR)**, which came into force on 25 May 2018. GDPR requires organisations to have a valid legal basis for processing personal data, to collect only what is necessary for a stated purpose ("data minimisation"), to allow individuals to access, correct or request erasure of their data, and to report data breaches. Because Ireland hosts the European headquarters of Meta, Google and other major platforms, Ireland's **Data Protection Commission (DPC)** is the lead GDPR regulator for these companies across the EU — giving this issue direct Irish relevance. The DPC has issued some of the largest GDPR fines to date against Meta, including a record €1.2 billion fine in 2023 over unlawful EU–US data transfers, and further fines running into the hundreds of millions of euro for breaches relating to data-processing design and breach notification — illustrating that these are not abstract concerns but ones with substantial, actively enforced legal consequences.

Implications for individual privacy and autonomy

Taken together, the ethical and legal picture points to the same underlying implication: individuals have progressively less practical control over their own data and, by extension, less control over the information environment that shapes the choices presented to them. Even where a legal framework such as GDPR grants individuals formal rights (access, correction, erasure, objection), those rights depend on the individual understanding what has been collected and successfully exercising them against a large, asymmetrically informed organisation — which is why critics such as Zuboff argue that surveillance capitalism is a systemic challenge to individual autonomy that legal reform alone has not fully resolved, rather than a problem solved simply by GDPR's existence.

> **SOURCING NOTE** *The definition of surveillance capitalism is attributed to Shoshana Zuboff (the term's originator); the GDPR effective date, and the DPC's role and specific fine figures against Meta Ireland, are drawn from the Irish Data Protection Commission's own published decisions rather than from IS1105 course material.*

Mark Allocation — Question 2(b)

Band	Descriptor	Marks
Excellent *21–25*	Correctly explains surveillance capitalism and connects it explicitly to individual privacy and autonomy; discusses genuine ethical concerns (consent, behavioural influence, power asymmetry) **and** legal concerns (correctly referencing GDPR, ideally with Irish/DPC relevance); well organised and critical rather than purely descriptive.	**21–25**
Good *16–20*	Reasonable explanation of surveillance capitalism with some ethical and legal discussion; GDPR referenced but less developed; link to autonomy present but underdeveloped.	**16–20**
Partial *10–15*	Basic or partial definition of surveillance capitalism; ethical or legal discussion present but not both; weak or missing link to privacy/autonomy implications.	**10–15**
Minimal *0–9*	Little or no accurate explanation of surveillance capitalism, or answer is generic/off-topic.	**0–9**

Question 2(b) Total: 25% | Question 2 Total: 50 Points